DAY 06 ASSIGNMENT

UNDERSTANDING IPv4 ADDRESSING

Submitted By: SREEKAKULAM THULASIRAM

Assignment Code: DAY06-IPV4

Topic: IPv4 Addressing and Network Configuration

1) IPv4 THEORY

1. What is an IPv4 address?

Internet Protocol version 4 (IPv4) is a network addressing protocol. This is used to identify devices on an IP network and allow the transfer of data packets. This is one of the most commonly used IP versions.

2. How many bits are there in an IPv4 address?

IPv4 addresses contain 32 bits. This is split into four sections of 8 bits.

3. What is an octet?

An Octet is a section of 8 bits. Within IPv4 addressing an octet can have a decimal value of 0-255.

4. How many octets are there in an IPv4 address?

There are four octets in an IPv4 address. They are separated by full stops. For example 192.168.1.10 has 4 octets. 192, 168, 1 and 10.

5. What is a subnet mask?

A subnet mask is a 32 bit number that is used to distinguish between the network and host components of an IPv4 address. This helps devices to determine if they are on the same local network as the target.

6. What is the difference between a network component and host component?

The network component identifies which specific network a device is on. The host component identifies which specific device is on the network.

A subnet mask is used to determine the network and host components of an IPv4 address.

7. What is a private IPv4 address?

A private IPv4 address is an address used within private networks such as home, office, colleges and companies. These cannot be routed onto the internet. They are used to allow devices within the private network to communicate with each other.

8. List all the private IPv4 address ranges.

1. 10.0.0.0 — 10.255.255.255
2. 172.16.0.0 — 172.31.255.255
3. 192.168.0.0 — 192.168.255.255

9. What is a public IPv4 address?

A public IPv4 address is an address used to identify devices or network interfaces on the public internet. These are unique globally and are designed to be routed.

10. What is the IPv4 loop back address?

The IPv4 loop back address is 127.0.0.1. This is used to allow the computer to communicate with itself. This can be used to test the local TCP/IP stack without sending packets to another device.

11. What is APIPA?

APIPA stands for 'Automatic Private IP Addressing'. It is a feature within Windows computers that allows them to automatically assign themselves an IPv4 link local address if they cannot get one from the DHCP server.

12. What does 169.254.x.x often mean?

If your IP address begins with 169.254.x.x then this often means that the computer has assigned itself an APIPA address. This normally occurs when the computer cannot get an IPv4 address from the DHCP server.

This could also indicate there is an issue with the DHCP server, network connection or router.

13. What is a default gateway?

A default gateway is usually a router. This is used to send data from the computer to another device that isn't on the local network.

14. What is the difference between Static and Dynamic IP addresses?

Static IP addresses are manually assigned or set as permanent for a device/network interface.

Dynamic IP addresses are automatically assigned to a device. Usually using DHCP. The IP address will generally have a lease time.

Static IP addresses are fixed. Dynamic IP addresses make it easier to manage your network.

15. What is DHCP?

DHCP stands for Dynamic Host Configuration Protocol. This automatically assigns network information to devices such as IP addresses, Subnet Masks, Default Gateway and DNS Server addresses.

This reduces the need to manually assign network settings.

16. What is a network address?

A network address is an IPv4 address that is used to identify a network/subnet. All host bits are turned to 0.

17. What is a broadcast address?

A broadcast address is an IPv4 address that sends data to all devices on a specific subnet. All host bits are turned to 1. All devices on the subnet will receive the broadcast.

18. Why is it important to know about IPv4?

Knowing about IPv4 can help security professionals understand how devices communicate on a network. Knowing about IP addresses, subnet masks, private/public IP ranges, gateways and DHCP can help you to identify devices on a network and investigate any suspicious activity.

Knowing about these can also help you identify unknown devices on your network, analyse network traffic, identify possible attacks and understand how attackers are communicating with systems. This can help security professionals to monitor and secure their networks.

2) PRACTICAL IDENTIFICATION

I used command prompt to look at the IPv4 settings on my computer. I used the command ipconfig and ipconfig /all.

I got the following information from my active network adapter:

Network Information	My Computer Information
IPv4 Address	10.47.230.97
Subnet Mask	255.255.255.0
Default Gateway	10.47.230.60
DNS Server	10.47.230.60
DHCP Enabled	yes

3) SCREENSHOTS

Screenshot 1 – IP Configuration

This screenshot shows the output when I used the command ipconfig in command prompt. It shows my IPv4 address, subnet mask and default gateway.

```
Wireless LAN adapter Wi-Fi 2:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Wireless-AC 9560 160MHz
   Physical Address. . . . . . . . . : 48-51-C5-A7-F6-66
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2409:40f0:5055:5886:6d5b:bc92:3ceb:3057(Preferred)
   Temporary IPv6 Address. . . . . . : 2409:40f0:5055:5886:ecf6:8807:6113:6ad(Preferred)
   Link-local IPv6 Address . . . . . : fe80::6b34:68ed:a853:a8f7%15(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.47.230.97(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Wednesday, 4 November 2026 7:27:55 pm
   Lease Expires . . . . . . . . . . : Wednesday, 4 November 2026 8:57:56 pm
   Default Gateway . . . . . . . . . : fe80::9435:4ff:fec2:a291%15
                                       10.47.230.60
   DHCP Server . . . . . . . . . . . : 10.47.230.60
   DHCPv6 IAID . . . . . . . . . . . : 155734469
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2B-10-43-5A-00-E0-4C-68-0A-A9
   DNS Servers . . . . . . . . . . . : 10.47.230.60
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

Screenshot 2 – IP Configuration (Detailed)

This screenshot shows the output when I used the command ipconfig /all. It shows detailed network information for my active network adapter. Including IPv4 address, subnet mask, default gateway, DNS server and DHCP.

```
PS C:\Users\Thulasiram> arp -a

Interface: 192.168.111.1 --- 0x7
  Internet Address      Physical Address      Type
  192.168.111.254       00-50-56-f7-01-11     dynamic
  192.168.111.255       ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 10.47.230.97 --- 0xf
  Internet Address      Physical Address      Type
  10.47.230.60          96-35-04-c2-a2-91     dynamic
  10.47.230.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

```
Wireless LAN adapter Wi-Fi 2:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Wireless-AC 9560 160MHz
   Physical Address. . . . . . . . . : 48-51-C5-A7-F6-66
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2409:40f0:5055:5886:6d5b:bc92:3ceb:3057(Preferred)
   Temporary IPv6 Address. . . . . . : 2409:40f0:5055:5886:ecf6:8807:6113:6ad(Preferred)
   Link-local IPv6 Address . . . . . : fe80::6b34:68ed:a853:a8f7%15(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.47.230.97(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Wednesday, 4 November 2026 7:27:55 pm
   Lease Expires . . . . . . . . . . : Wednesday, 4 November 2026 8:57:56 pm
   Default Gateway . . . . . . . . . : fe80::9435:4ff:fec2:a291%15
                                       10.47.230.60
   DHCP Server . . . . . . . . . . . : 10.47.230.60
   DHCPv6 IAID . . . . . . . . . . . : 155734469
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2B-10-43-5A-00-E0-4C-68-0A-A9
   DNS Servers . . . . . . . . . . . : 10.47.230.60
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Hyper-V Virtual Ethernet Adapter
   Physical Address. . . . . . . . . : 00-15-5D-61-18-A5
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::2598:9ba0:9568:cb8%45(Preferred)
   IPv4 Address. . . . . . . . . . . : 172.27.16.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 754980189
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2B-10-43-5A-00-E0-4C-68-0A-A9
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

Screenshot 3 – Loopback Test

This screenshot shows the output when I used the command **ping 127.0.0.1**. It shows the loopback test I did to test if my TCP/IP stack is responding correctly.

```
PS C:\Users\Thulasiram> ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\Users\Thulasiram>
```

CONCLUSION

Throughout this assignment I have learnt about IPv4 addressing, octets, subnets masks, private and public IP addresses and different types of IP configurations. I also learnt what APIPA, DHCP, default gateways, network addresses and broadcast addresses do. Through the practical tasks I have learnt how to use the commands ipconfig, ipconfig /all and ping 127.0.0.1 to check my network settings on my computer and to see how my local network works. This can be useful for understanding how networks work and how security professionals can identify and secure devices on a network.